



Recent applications of magnesium- and Zinc-TMP amides in the synthesis of bioactive targets

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ABSTRACT

Magnesium- and zinc-TMP amides have come out from relative obscurity to be recognized as important tools in modern synthetic chemistry. Development of LiCl-activated metal amides like $\text{TMPMgCl} \cdot \text{LiCl}$ and $\text{TMP}_2\text{Zn} \cdot 2\text{MgCl}_2 \cdot 2\text{LiCl}$ has been important to obtain new organometallic reagents through highly chemo- and regioselective reactions. Herein, we review how these reagents are applied in the synthesis of important bioactive targets such as natural products and drugs.

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1. Introduction

Using metal amides to transform a relatively inert C–H bond into a more reactive C–M bond selectively is a key strategy in organic synthesis [1,2]. For this purpose, hindered lithium amides, such as lithium diisopropylamide (LDA) and lithium 2,2,6,6-tetramethylpiperidide (LiTMP), are very useful bases that allow several aliphatic and aromatic substrates to be regioselectively deprotonated [2]. However, the high ionic character of the C–Li bond may preclude organolithium reagents from being applied in substrates bearing important functional groups, like ester and cyano.

Over the last years, magnesium amides, first reported by Hauser in 1947 [3], have emerged as relevant alternatives to lithium amides because the corresponding magnesium Grignard reagents display superior thermal stability and higher tolerance to the presence of sensitive functional groups [4]. In 1989, Eaton reported that

directed ortho-magnesiations of aromatic substrates like methyl benzoate could be conducted at room temperature by using Hauser bases [5]. Mulzer [6] and Kondo [7] further expanded the scope of this methodology to functionalized heterocyclic substrates including pyridines, thiazoles, and indoles. Moreover, chiral magnesium amides can be potentially applied in enantioselective deprotonation reactions [8].

Despite the potential applications of Hauser bases in organic synthesis, these compounds tend to aggregate and hence be poorly soluble in commonly used ethereal solvents. Therefore, large excess of the magnesium amide may be necessary to achieve full metalation. Knochel and co-workers circumvented this problem by developing LiCl-activated TMP bases. The mixed Li/Mg amides $\text{TMPMgCl} \cdot \text{LiCl}$ (**1**) [9] and $\text{TMP}_2\text{MgCl} \cdot 2\text{LiCl}$ (**2**) [10] proved to be more soluble and reactive reagents and were able to promote highly selective magnesiation in the presence of stoichiometric amounts of the base. Reagents **1** and **2** have complementary reactivities; amide $\text{TMPMgCl} \cdot \text{LiCl}$ is largely used for magnesiation of reactive heterocyclic substrates, while diamide $\text{TMP}_2\text{MgCl} \cdot 2\text{LiCl}$ is normally applied to magnesiate moderately activated substrates such as *tert*-butyl benzoate (Fig. 1) [9,10].

In addition, the Li/Zn amides $\text{TMP}_2\text{Zn} \cdot 2\text{MgCl}_2 \cdot 2\text{LiCl}$ (**3**) [11] and

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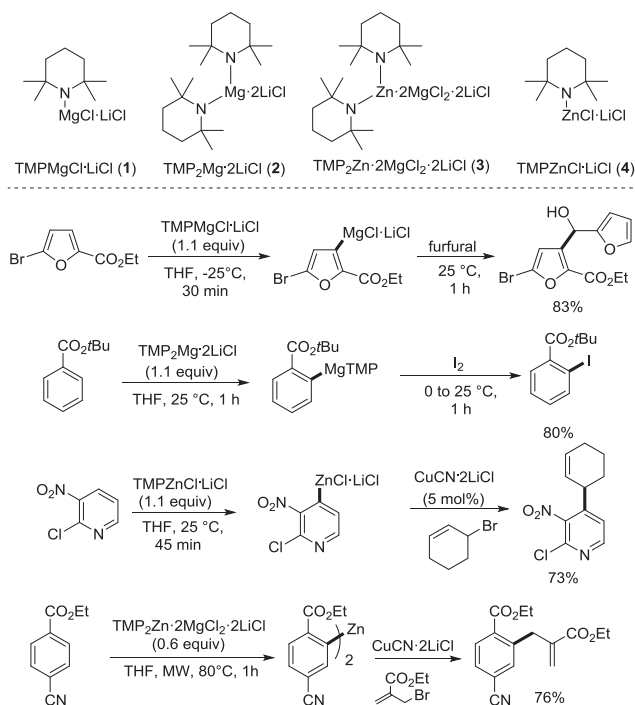


Fig. 1. Structure and some applications of Mg/Li, Zn/Mg/Li and Zn/Li TMP-amides.

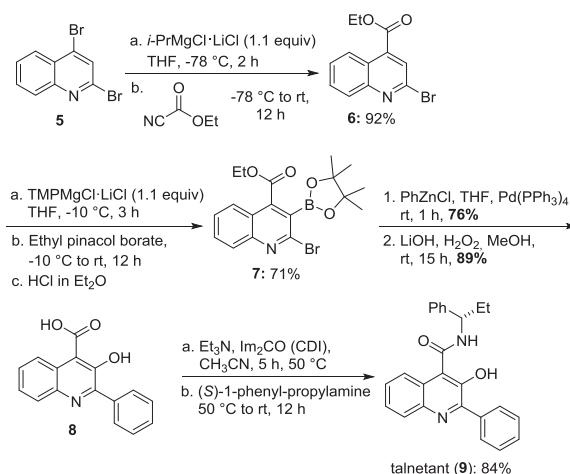
TMPZnCl·LiCl (**4**) [12] may provide selective zincation at room temperature or even under heating (including microwave heating) in very sensitive aromatic substrates in the presence of reactive functional groups (Fig. 1) [11,12]. Extensive investigations into these mixed metal amides have shown their wide applicability in the chemo- and regioselective metalation of numerous arenes and heteroarenes [13]. Furthermore, since both magnesium and zinc have two empty low-lying *p*-orbitals, they can be transmetalated with transition metals (e.g., palladium, copper, or nickel), leading to intermediates containing *d*-orbitals, which allows coupling with various electrophiles [14].

To bring this exciting class of reagents to attention, we aim to present a brief review of the recent applications of magnesium- and zinc-TMP amides in the synthesis of bioactive targets, such as natural products and drugs.

2. Synthetic applications of magnesium-TMP amides

After investigating multiple regioselective polybrominated quinoline magnesiations, Knochel and co-workers applied the developed methodology to synthesize talnetant (**9**) [15], a selective NK-3 antagonist developed by GlaxoSmithKline [16]. Regioselective bromo-magnesium exchange between 2,4-dibromoquinoline (**5**) and *i*-PrMgCl·LiCl (turbo-Grignard) [17] followed by reaction with ethyl cyanofornate afforded 4-carbomethoxy-2-bromoquinoline (**6**) as the sole product in 92% yield. In the next step, 1.1 equiv. of TMPMgCl·LiCl was used for directed magnesiation at position 3. The corresponding organomagnesium intermediate readily reacted with ethyl pinacol borate, to give pinacol boronic ester **7** in 71% yield. Subsequently, a sequence of Negishi cross-coupling, basic oxidation, and amide formation furnished talnetant (**9**) in good overall yield (Scheme 1).

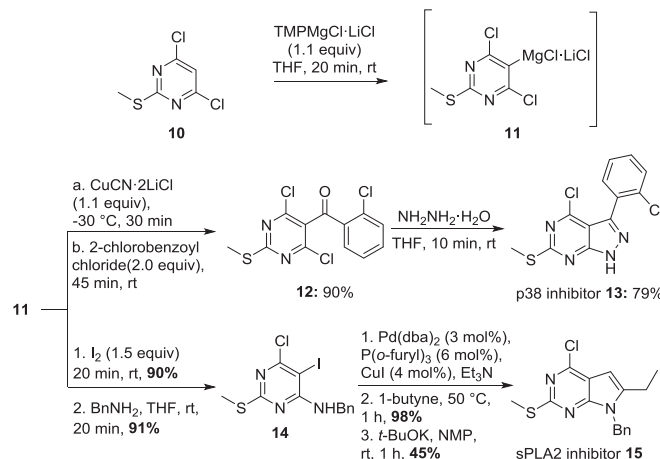
The pyrimidine scaffold exists in the structure of relevant bioactive compounds. In 2008, Mosrin and Knochel studied pyrimidine magnesiation with TMPMgCl·LiCl [18]. They also employed this strategy to prepare inhibitors of protein kinase p38



Scheme 1. Total synthesis of talnetant (**9**).

and phospholipase sPLA2, which are antiviral and anti-inflammatory targets [19]. In both syntheses, 4,6-dichloro-2-(methylthio)pyrimidine (**10**) was used as starting material. Interestingly, although lithiation of **10** with LDA had been reported at $-78\text{ }^{\circ}\text{C}$ for 3 h [20], magnesiation with TMPMgCl·LiCl took place at room temperature within 20 min. Transmetalation of organo-magnesium reagent **11** with CuCN·2LiCl followed by acylation with 2-chlorobenzoyl chloride gave ketone **12** in 90% yield. Subsequent cyclization with $\text{NH}_2\text{NH}_2\cdot\text{H}_2\text{O}$ provided the p38 kinase inhibitor **13** in 79% yield. On the other hand, iodolysis of **11** followed by amination at position 6 with benzylamine furnished pyrimidine **14** in 82% yield (two steps). After a Sonogashira cross-coupling with 1-butyne, *t*-BuOK-mediated cyclization of the corresponding amino-substituted aryl-alkyne gave phospholipase sPLA2 inhibitor **15** in reasonable yield (Scheme 2).

Direct lithiation of pyrimidines is known to be difficult due to high reactivity of the ring toward addition of organolithium reagents [21]. In this context, magnesiation appears as an interesting option. In 2009, Mosrin and Knochel [22] used the base TMPMgCl·LiCl in two sequential steps to obtain the fungicide mepanipyrim [23]. Regioselective di-functionalization of 2-chloropyrimidine (**16**) in the presence of iodine and 1,2-dibromotetrachloroethane as electrophiles afforded 4-bromo-2-chloro-6-iodopyrimidine (**18**) in 87% overall yield. Next, a Negishi



Scheme 2. Synthesis of the p38 (**13**) and sPLA2 (**15**) inhibitors.

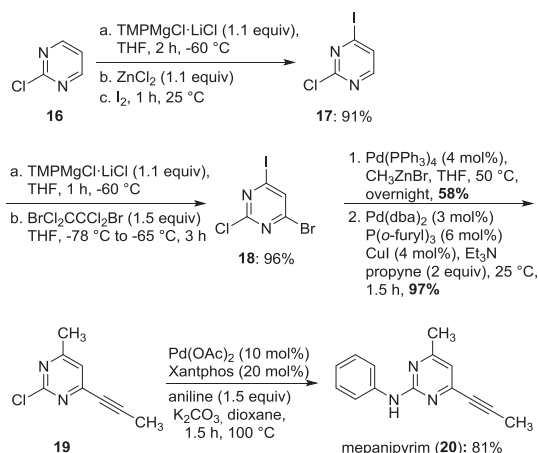
cross-coupling with CH_3ZnBr followed by a Sonogashira reaction with 1-butyne allowed arylalkyne **19** to be isolated in 56% yield (two steps). Finally, a Buchwald-Hartwig amination with aniline furnished the bioactive target **20** in 81% yield (Scheme 3).

Pyrazines are important *N*-heterocycles present in the structure of numerous naturally occurring bioactive compounds. 2,3-Disubstituted pyrazines can be obtained by regioselective chloropyrazine lithiation followed by reaction with electrophiles [24]. In 2009, Mosrin and Knochel showed that chloropyrazine derivatives could be more efficiently functionalized with the aid of mixed Li/Mg- and Li/Zn-TMP amides [25]. Synthesis of coelenterazine (**25**), an important light-emitting compound found in many marine organisms, illustrated the relevance of this methodology [26]. After selective metalation of **21** with $\text{TMPMgCl}\cdot\text{LiCl}$ and transmetalation with ZnCl_2 , a palladium-catalyzed acylation with benzoyl chloride afforded ketone **22** in 71% yield. Interestingly, amination with NH_3 /butanol, methyl ether cleavage with sodium ethanethiolate, and a Wolf-Kishner reduction led to coelenteramine **23** in 66% (three steps). Condensation of **23** with the 2-oxo-3-phenylpropanal derivative **24** allowed **25** to be isolated in 64% yield (Scheme 4).

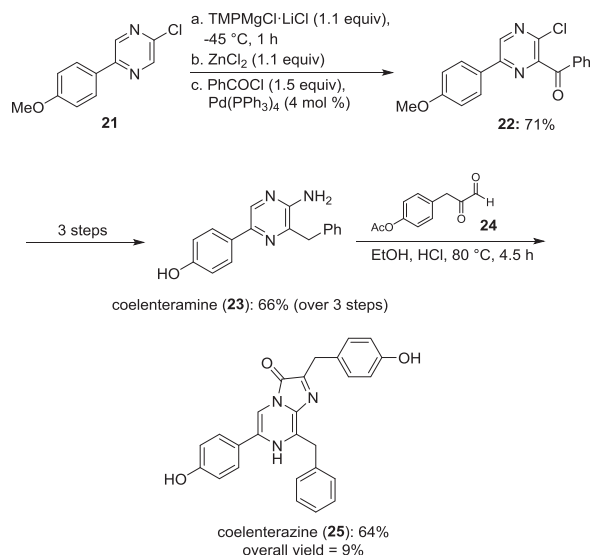
In 2003, Fusetani's group isolated five new alkaloids (dictyodendrins A-E) from the marine sponge *Dictyodendrillum verongiformis* (Fig. 2). This new class of marine alkaloids displayed noteworthy inhibitory activity against telomerase, an enzyme found to be normally expressed in 90% of human malignant tumors [27].

Remarkably, in 2011 Tokuyama and co-workers reported on the synthesis of all five alkaloids by using a highly substituted indole (**30**) as a common intermediate [28]. To synthesize the indoline precursor **29** from substrate **27**, the magnesium diamide $\text{TMP}_2\text{Mg}\cdot 2\text{LiBr}$ [29] was more efficient than TMPLi and other hindered bases in the planned benzyne generation-cyclization sequence. In this context, a relevant limitation of the use of TMPLi as a base was formation of a TMP adduct with rising temperature. In contrast, reaction of **27** with $\text{TMP}_2\text{Mg}\cdot 2\text{LiBr}$ allowed effective preparation of magnesiated indoline **2**. Moreover, transmetalation of this intermediate with CuI was crucial for arylation to occur in high yield (Scheme 5).

Tokuyama group also employed the indoline synthesis strategy to prepare important pyranindoles [30], which have potential activity against hepatitis C virus (HCV) [31]. A benzyne-mediated cyclization was in this case performed by $\text{TMP}_2\text{Mg}\cdot 2\text{LiCl}$ followed by the corresponding organocopper intermediate methylation with iodomethane provided the key indoline **32** in 80% yield (Scheme 6). These results indicate that there is no significant difference in using $\text{TMP}_2\text{Mg}\cdot 2\text{LiBr}$ or $\text{TMP}_2\text{Mg}\cdot 2\text{LiCl}$ bases in these reactions, since



Scheme 3. Total synthesis of mepanipyrim (**20**).



Scheme 4. Total synthesis of coelenterazine (**25**).

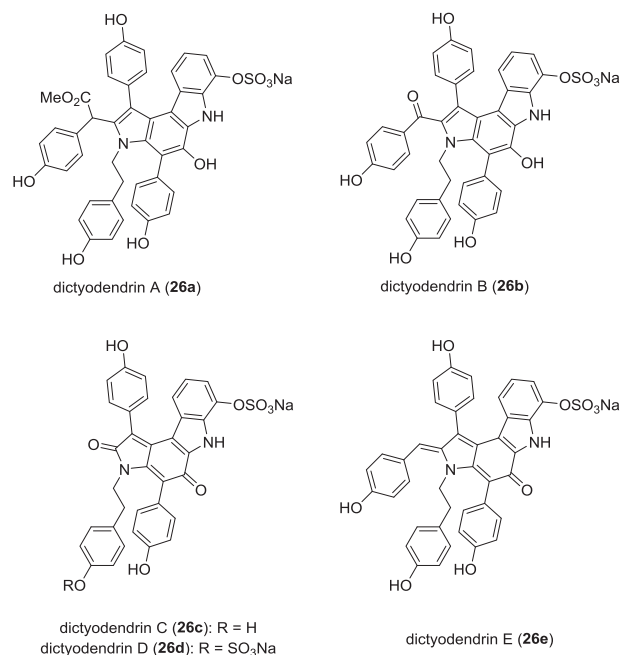
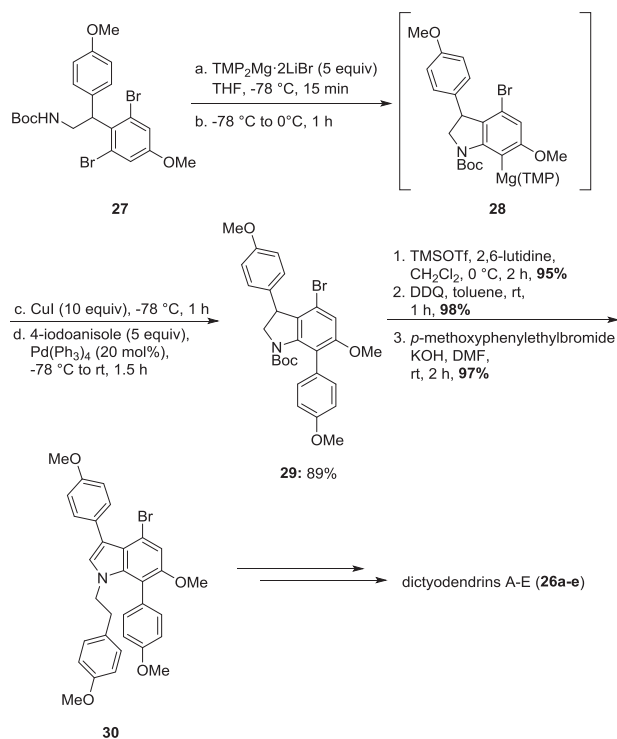
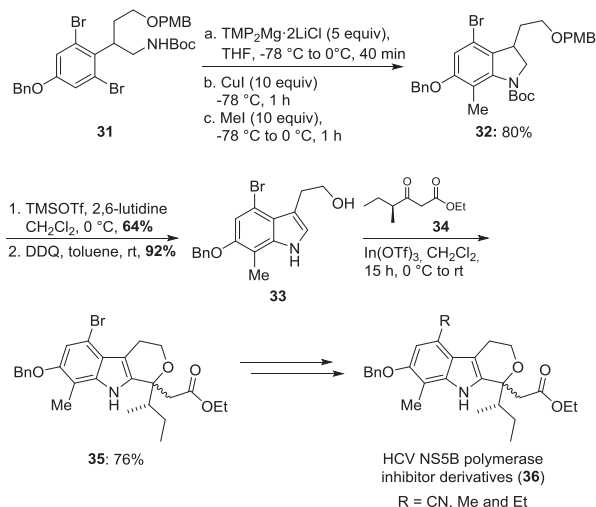


Fig. 2. Dictyodendrins A-E (**26a-e**).

both indolines **29** and **32** were obtained in similar yields when 5 equivalents of the corresponding reagents were used.

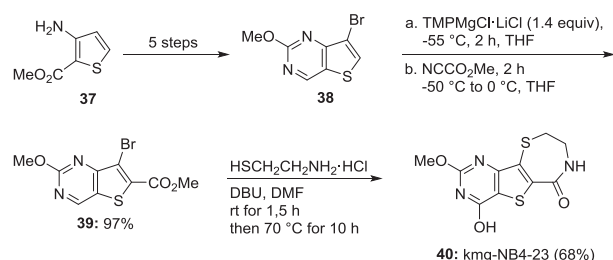
After Boc and PMB groups were removed, $\text{In}(\text{OTf})_3$ catalyzed the oxa-Pictet-Spengler reaction of **33** with ketoester **34**, to give pyranindole **35** in 76% yield. Further synthetic manipulation allowed cyano, methyl, and ethyl groups to be introduced at C-5 in the pyranindole ring (Scheme 6).

In 2011, Wipf and co-workers described the synthesis and biological evaluation of novel benzothienothiazepinone analogues that can inhibit protein kinase D [32]. The authors synthesized a promising lead structure named kmg-NB4-23 (**40**) from methyl 3-aminothiophene-2-carboxylate (**37**) in seven steps. Notably, selective magnesiation of **38** with $\text{TMPMgCl}\cdot\text{LiCl}$ at C-6 followed by quenching with methyl cyanofomate allowed the methyl ester

Scheme 5. Synthesis of an important intermediate (**30**) of dictyodendrins A-E (**26a-e**).Scheme 6. Preparation of three HCV NS5B RNA polymerase inhibitor (**36**) derivatives.

derivative **39** to be isolated in 97% yield. To build the thiazepinone moiety, a one-pot bromide displacement-condensation of **39** with cysteamine hydrochloride was performed, which afforded **40** in 68% yield (Scheme 7).

In 2004, Singh and co-workers isolated three new 11-membered unsaturated lactones from *Aspergillus* sp. Among these molecules, aspercyclides A inhibited immunoglobulin E (IgE) binding to receptors with an IC_{50} of $200\ \mu\text{M}$ [33]. Recently, Sato and co-workers achieved total synthesis of aspercyclides A and B when they used an interesting intramolecular oxidative C–O bond formation in the key step [34]. Interestingly, the authors obtained two important synthetic intermediates of the natural products from dioxinone **42**, which they had readily prepared in 83% yield through *ortho*-

Scheme 7. Synthesis of kmg-NB4-23 (**40**).

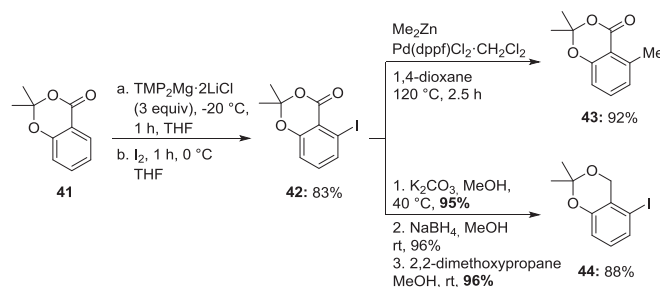
metalation of dioxinone **41** with $\text{TMP}_2\text{Mg}\cdot 2\text{LiCl}$ as selective base and iodine as electrophile (Scheme 8).

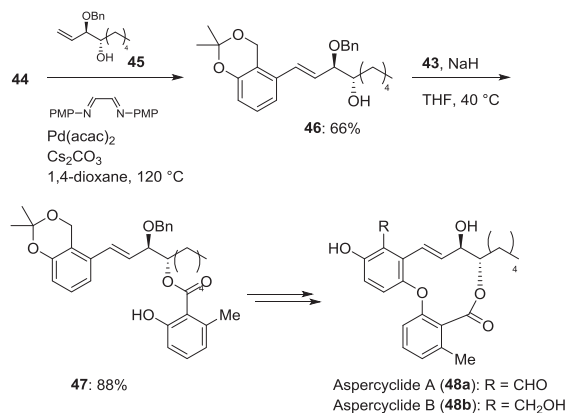
A Mizoroki-Heck reaction of iodide **44** with alkene **45** furnished **46** in 66% yield. Reaction of alcohol **46** with dioxinone **43** gave ester **47** in 88% yield, which was further converted to Aspercyclides A and B after several synthetic steps (Scheme 9).

In 2011, discovery of various novel non-ATP linear competitive protein kinase MK2 inhibitors was reported [35]. The MK2 cascade directly regulates tumor necrosis factor α ($\text{TNF}\alpha$) production, thus anti-TNF is used to treat autoimmune diseases such as rheumatoid arthritis, Crohn's disease, and psoriasis [36]. In 2013, Xiao and co-workers synthesized new tricyclic MK2 inhibitors of type **53** in a few steps [37]. The synthetic approach involved directed *ortho*-metalation of the commercially available furans **49**. In this case, a limitation to the use of lithium bases would be low tolerance of organolithium reagents to the presence of ethyl ester groups. Notably, reactions of **49** with $\text{TMPMgCl}\cdot\text{LiCl}$ followed by quenching of the organomagnesium intermediates with aldehyde **50** afforded the expected alcohols of type **51** in 60–70% yields. Subsequent reduction of nitro and hydroxyl groups, lactamization, *N*-alkylation, and deprotection produced the desired tricyclic lactams of type **53** in reasonable isolated yields (Scheme 10).

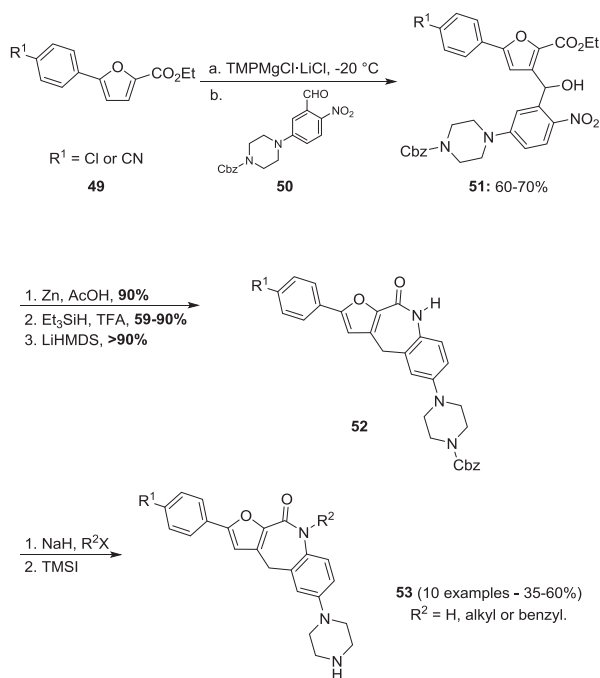
Numerous marine invertebrates, especially sponges and ascidians, are often brightly colored due to the presence of pyridoacridine alkaloids [38]. In 2014, Bracher and co-workers synthesized a marine pyridoacridine alkaloid named demethyldeoxyamphimedine (**60**) [39], which was isolated from the marine ascidian *Cystodytes dellechiaiei* in 2011 [40].

The synthetic route started with frustrated Lewis pairs $\text{TMPMgCl}\cdot\text{BF}_3\cdot\text{LiCl}$ in the selective functionalization of ethyl nicotinate (**54**). This reagent, easily prepared by mixing the base $\text{TMPMgCl}\cdot\text{LiCl}$ with $\text{BF}_3\cdot\text{OEt}_2$, can perform regioselective C-4 functionalization of pyridines bearing functional groups at C-3 [41]. After transmetalation with ZnCl_2 , a palladium-catalyzed Negishi cross-coupling of **55** with 2-iodoaniline (**56**) formed a biaryl intermediate, which provided product **57** in 50% yield after intramolecular lactamization. Compound **57** was then converted into

Scheme 8. Use of $\text{TMP}_2\text{Mg}\cdot 2\text{LiCl}$ in the synthesis of aspercyclides A and B intermediates **43** and **44**.



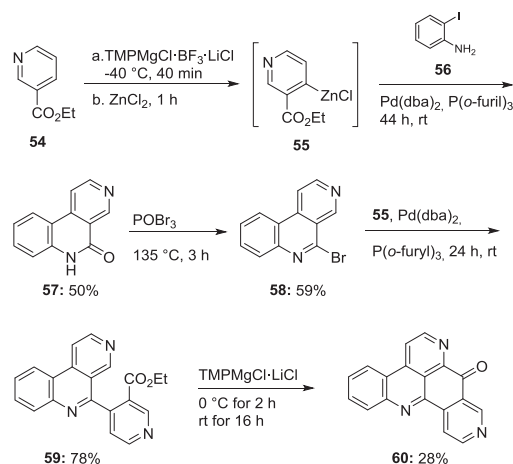
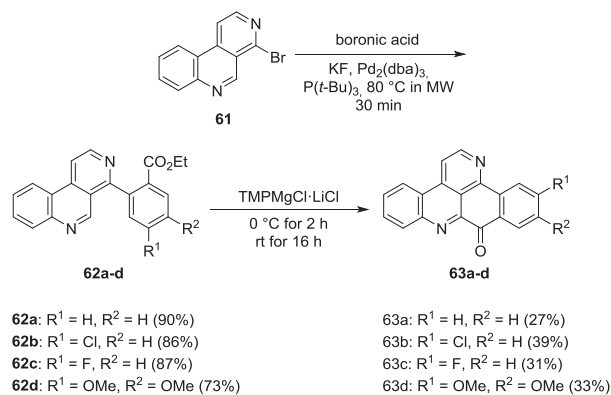
Scheme 9. Synthesis of Aspercyclides A and B.

Scheme 10. Synthesis of new tricyclic MK2 inhibitors (**53**).

aryl-bromide by using POBr₃ and submitted to a Negishi cross-coupling with organozinc **55**, to generate **59** in 78% yield. Finally, highly controlled metalation of **59** with TMPMgCl·LiCl followed by intramolecular cyclization allowed the desired marine natural product **60** to be isolated in 28% yield (Scheme 11).

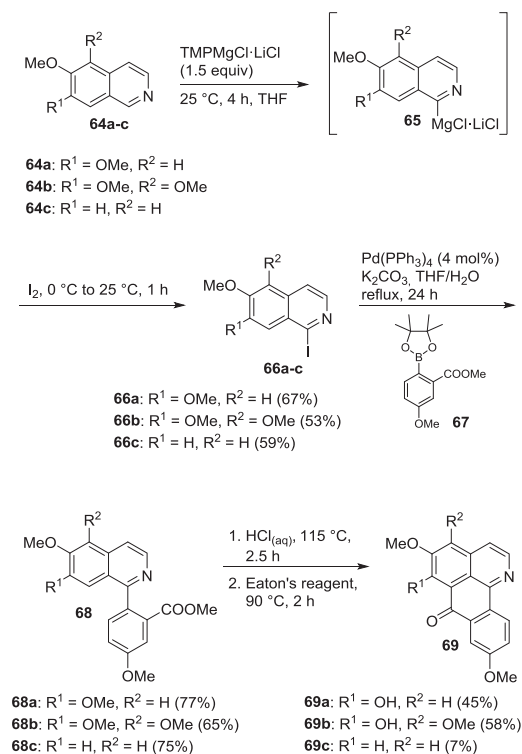
In 2015, the same group synthesized various pyridoacridine alkaloids by a similar methodology [42]. Suzuki cross-coupling reactions between 4-bromobenzo[c][2,7]naphthyridine (**61**) and boronic acids led to four derivatives of type **62** with yields ranging from 73% to 90%. Furthermore, treatment of ester **62** with TMPMgCl·LiCl followed by intramolecular cyclization gave pentacyclic compounds **63a-d** with yields varying from 27% to 33% (Scheme 12).

More recently, Bracher's group also developed a new strategy to prepare some benzyloisoquinoline alkaloids [43], which were isolated from *Menispermum dauricum* DC rhizome [44]. The authors used TMPMgCl·LiCl base to metalate alkoxy isoquinolines **64a-c** regioselectively. After attempts of using the Negishi reaction to

Scheme 11. Synthesis of demethyldeoxyamphimedine (**60**).Scheme 12. Synthesis of Deazaascididemin (**63a**) and analogues **63b-d**.

reach the biaryl derivatives of type **68** failed, the authors quenched **65a-c** with iodine, to obtain the iodoisoquinolines of type **66** (53–67% yields), which they then readily converted into **68a-c** (65–77% yields) by Suzuki cross-couplings in the presence of boronic acid pinacol ester **67**. Further acid-catalyzed ester hydrolysis followed by cyclization with Eaton's reagent allowed 6-O-demethylmenisporphine (**69a**), dauriporphinoline (**69b**), and bianfugene (**69c**) to be isolated from poor to reasonable yields (Scheme 13).

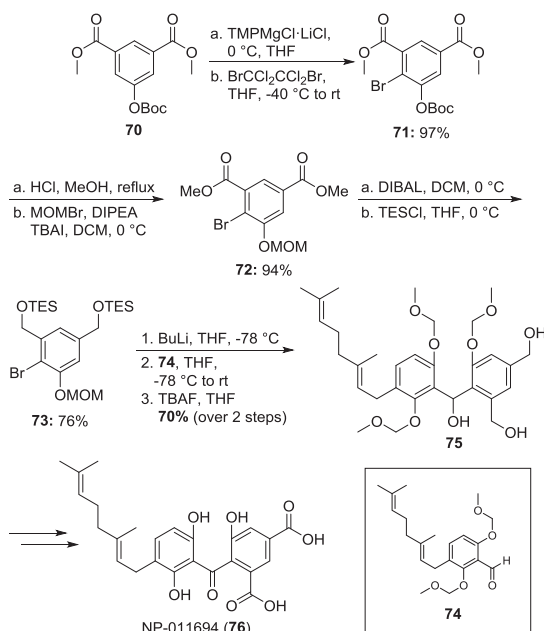
Recently, the search for new non-nutritive sweeteners has attracted much attention. In this context, Noncovich and co-workers employed the hT1R2/hT1R3 cell-based assay to screen various commercially available natural products [45]. Among these compounds, NP-011694 (**76**) presented high potential for such purpose, so the authors investigated its total synthesis. To achieve a brominated building block **73**, the authors initiated the synthesis by carrying out selective magnesiation of compound **70** with TMPMgCl·LiCl. After quenching with 1,2-dibromotetrachloroethane, product **71** was obtained in 97% yield. This result illustrated the superior tolerance of organomagnesium reagents to the presence of functional groups in aromatic substrates. Subsequently, a set of deprotection, protection, and reduction reactions provided intermediate **73**. After a bromine-lithium exchange reaction of **73** with *n*-BuLi, the corresponding organolithium derivative was reacted with aldehyde **74** and then treated with tetrabutylammonium fluoride to afford **75** in 70% yield over two steps, which was then converted into NP-011694 in three



Scheme 13. Synthesis of benzyloisoquinoline alkaloids (69a-c).

steps (Scheme 14).

In 2003, Fusetani and co-workers isolated four active compounds from marine invertebrates of the *Didemnum proliferum* species, including enediyne compound shishijimicin A, which exhibited highly potent antitumor properties [46]. Nicolaou and co-workers then performed the first total synthesis of this molecule [47]. In a convergent synthesis, the authors obtained the building block **78** by metalating carboline **77** with mixed lithium/

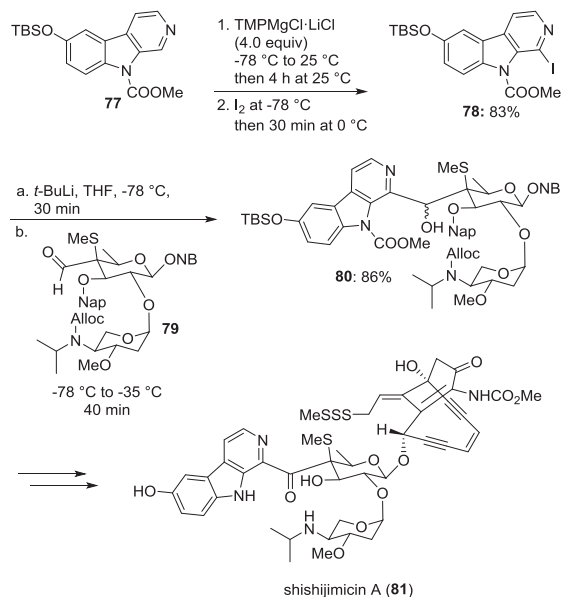
Scheme 14. Synthesis of the building block **73**.

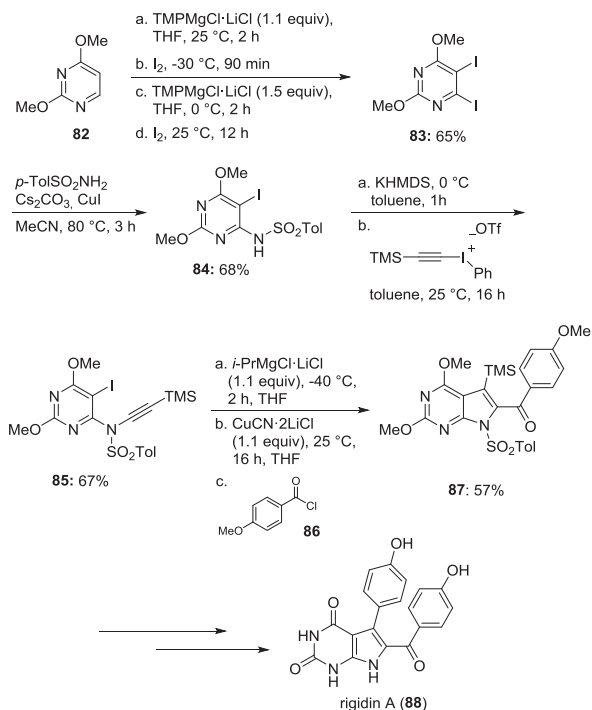
magnesium amide TMPMgCl·LiCl, followed by quenching with I₂. Lithium-iodide exchange with *t*-BuLi allowed coupling with disaccharide aldehyde **79** with 86% yield. Shishijimicin A (**81**) was then obtained after several steps (Scheme 15).

In 2016, Knochel's group demonstrated the applicability of mixed lithium/magnesium amides to prepare marine alkaloid rigidin A (**88**) [48], which has potent calmodulin antagonistic action; 2,4-dimethoxypyrimidine (**82**) was used as starting material [49]. Although lithiation of **82** with LiTMP at 0 °C has been reported [50], the use of TMPMgCl·LiCl as a base and iodine as electrophile allowed sequential double functionalization of the substrate to give the diiodo derivative **83** in 65% isolated yield.

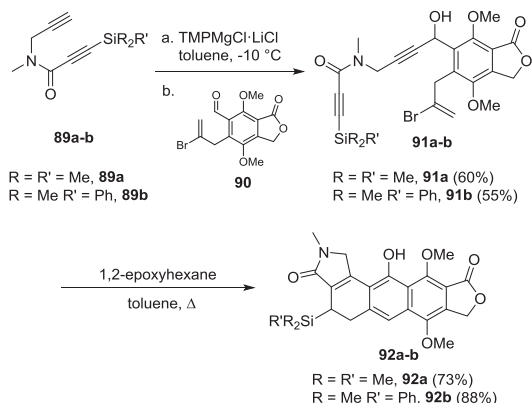
After copper-catalyzed amination of **83**, deprotonation of the resulting pyrimidyl *N*-sulfonylamide **84** with KHMDS and reaction with phenyl(trimethylsilyl)ethynyl iodide afforded the corresponding *N*-alkynylation product **85** in 67% yield. Remarkably, following the I-Mg exchange promoted by the turbo Grignard reagent, the organomagnesium intermediate was transmetalated with CuCN·LiCl, to afford pyrrolo[2,3-*d*]pyrimidine **87** in 57% yield after carbocupration reaction and benzoylation with **86**. Rigidin synthesis was completed after Suzuki arylation at C-3 and *N*-sulfonylamide and methoxy groups cleavage (Scheme 16).

In 1996, Matsumoto and co-workers isolated the potent antibiotic agent lactonamycin from *Streptomyces rishiriensis* [51]. Seven years later, Fiedler isolated lactonamycin Z from *Streptomyces sanglieri* [52], and this agent also presented antitumor activity. One of the greatest challenges regarding the preparation of this kind of compound is related to a pentacyclic core [53]. Recently, Parsons and co-workers developed a new approach to synthesize the core of this important class of compounds through a free-transition-metal cyclization strategy [54]. Aiming at the synthesis of alcohols of type **91**, first the authors investigated addition of diyne **89a** to aldehyde **90** in the presence of ZnBr₂. However, under these conditions, the zincated alkyne did not emerge, and a naphthalene derivative of **90** was isolated as sole product. In contrast, the corresponding magnesium acetylides were easily generated by reactions between alkynes **89a-b** and TMPMgCl·LiCl to give alcohols **91a-b** in 60% and 55% yield, respectively. Lastly, thermal cyclization in medium containing toluene and 1,2-epoxyhexane furnished products **92a** (73%) and **92b** (88%) in good yields (Scheme 17).

Scheme 15. Total synthesis of Shishijimicin A (**81**).



Scheme 16. Synthesis of alkaloid 88.

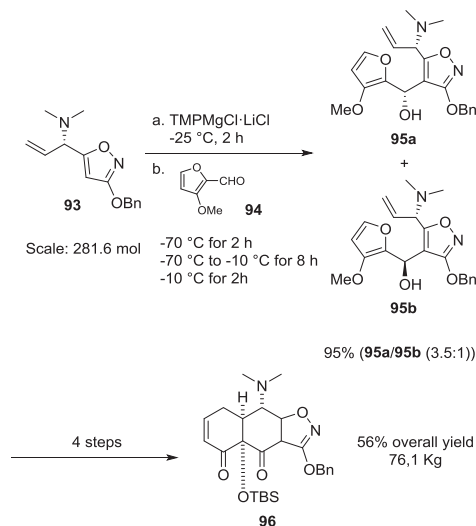


Scheme 17. New approach to synthesize the lactonamycin core (92a-b).

Recently, Zhang and co-workers synthesized enone 96, which is an important precursor for tetracycline antibiotics [55], by metalation of isoxazole ring 93 in the key step [56]. First, the group screened for metalation conditions by means of lithium reagents *n*-BuLi, *n*-hexyllithium, and LiTMP. However, very low temperatures were required, and sub-optimal results were obtained, with allylic metalation competing for the isoxazole deprotonation. Such issues were alleviated with the use of TMPMgCl·LiCl. After reacting with furaldehyde 94, the authors obtained carbinols 95a-b in 95% yield (95a/95b (3.5:1)). Interestingly, this reaction was successfully performed on a large scale (281.6 mol) (Scheme 18).

3. Synthetic applications of zinc-TMP amides

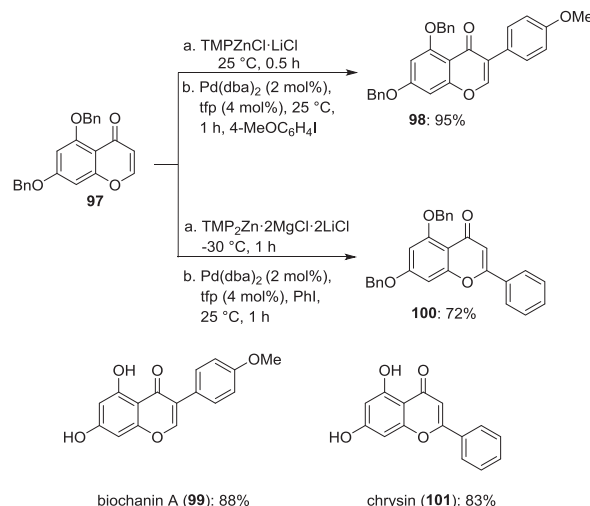
Previous literature reports have indicated how difficult it is to functionalize chromones directly by using lithium bases. Normally, lithiation requires very low temperatures, and the products are



Scheme 18. Preparation of key tetracycline precursor 96.

obtained in only modest yields after reaction with electrophiles [57]. In 2012, Knochel and co-workers showed that TMPZnCl·LiCl and TMP₂ZnCl₂·2LiCl could be efficiently used in the regioselective functionalization of chromones, thiochromones, and *N*-methyl-4-quinolones [58]. Notably, the presence of Lewis acids such as MgCl₂ and BF₃·OEt controls metalation at C-2, whereas TMPZnCl·LiCl performs selective metalations at C-3 in those substrates. These complementary methodologies were applied to synthesize biochanin A (99) [59] and chrysin (101) [60] starting from the same substrate. Selective zincation at C-3 in 97 with TMPZnCl·LiCl followed by a Negishi cross-coupling reaction with iodobenzene led to 98 in 95% yield. On the other hand, metalation of 97 with TMP₂ZnCl₂·2MgCl₂·2LiCl followed by a Negishi cross-coupling with 4-iodoanisole afforded the 2-aryl derivative 100 in 72% yield. Further hydrogenolysis of the corresponding benzyl groups allowed the natural products 99 and 101 to be isolated in 88% and 83% yield, respectively (Scheme 19).

In 2013, Carrera et al. synthesized a lead compound that inhibited PI3Kδ activity [61]. Recent studies have pointed to the key role of this specific kinase in rheumatoid arthritis progression,



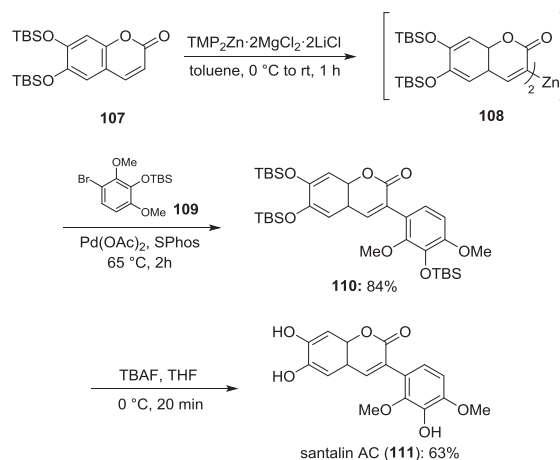
Scheme 19. Preparation of the chromones biochanin A (99) and chrysin (101).

making PI3K δ a potential target to treat this autoimmune disease [62]. Preparation of one representative compound of known PI3K δ inhibitors, GNE-026 (**106**), was possible through a TMPZnCl·LiCl-mediated metalation step. After scanning for optimum metalation/halogenation protocols, the authors found that the best condition was to employ TMPZnCl·LiCl to metalate **102**, followed by quenching with iodine solution, to give **103** in 86% yield on a preparative scale (140 g). By using TMPZnCl·LiCl, the authors also overcame low reactivity issues when purine substrate of type **102** reacted with bases like *n*-BuLi, LDA, and TMPMgCl·LiCl in the presence of ZnCl₂. Thereafter, etherification with *N*-Boc-azetidinol (**104**), Boc group deprotection with HCl, and reductive amination with 1,1-dioxothian-4-one and sodium triacetoxyborohydride provided GNE-293 (**106**) in 41% overall yield (Scheme 20).

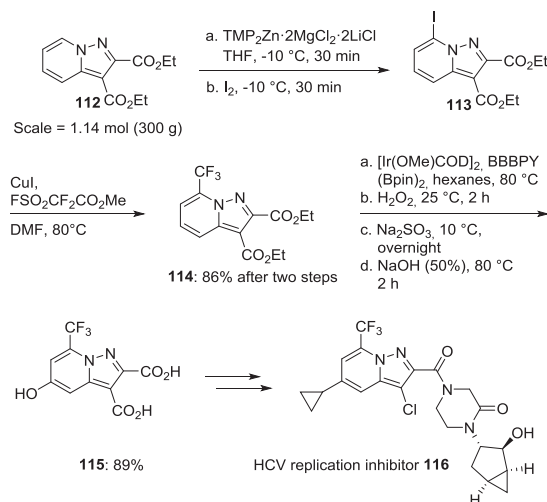
In 2013, Trauner and Strych performed the first total synthesis of santalins A, B, and AC as well as santarubins A and B [63], which are red sandalwood pigments isolated from the tree *Pterocarpus santalinus* [64]. In the synthesis of santalin AC (**111**), selective zincation of the coumarin **107** with TMP₂Zn·2MgCl₂·2LiCl followed by palladium-catalyzed cross-coupling reaction with aryl bromide **109** led to isoflavonoid **110** in 84% yield. Further TBAF silyl ether deprotection of **110** allowed natural product **111** to be isolated in 63% yield (Scheme 21).

Selective deprotonation by zinc-TMP bases was also shown to be useful to prepare important antiretroviral drug core scaffolds. Failing to use lithium and magnesium bases in a metalation step, researchers at the GSK R&D department successfully employed TMP₂Zn·2MgCl₂·2LiCl to obtain trifluoromethylated pyrazolopyridine **114** in 86% yield through selective zincation/iodation of **112** followed by Cu-mediated trifluoromethylation of intermediate **113** (Scheme 22) [65]. Iridium-catalyzed activation resulted in functionalization at C-5 in the pyrazolopyridine core [66], and the synthesis of **116**, a promising hepatitis C virus (HCV) inhibitor [67], was complete after several steps (Scheme 22).

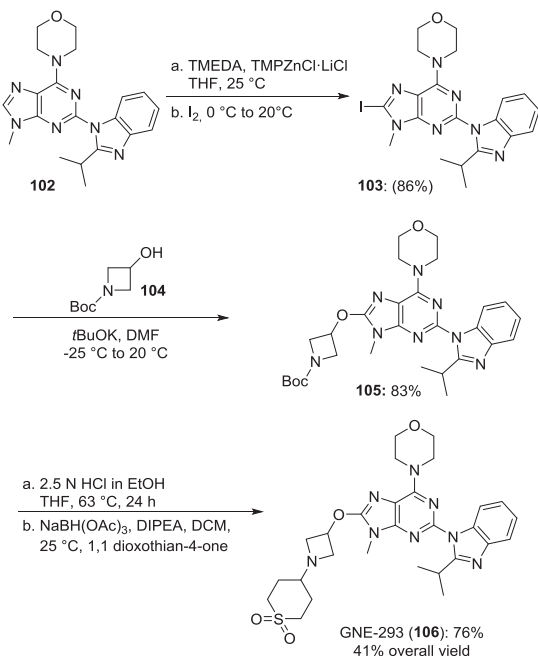
In 2016, Liu and Wang conducted arylation, vinylation, and alkynylation of numerous (hetero)arenes by zincation in the presence of TMPZnCl·LiCl followed by copper (I) triflate-catalyzed



Scheme 21. Synthesis of santalin AC (**111**).



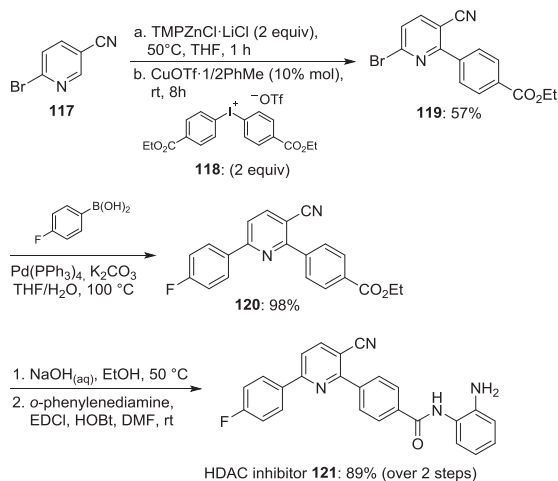
Scheme 22. Preparation of the antiretroviral precursor **115**.



Scheme 20. Synthesis of the PI3K δ inhibitor GNE-293 (**106**).

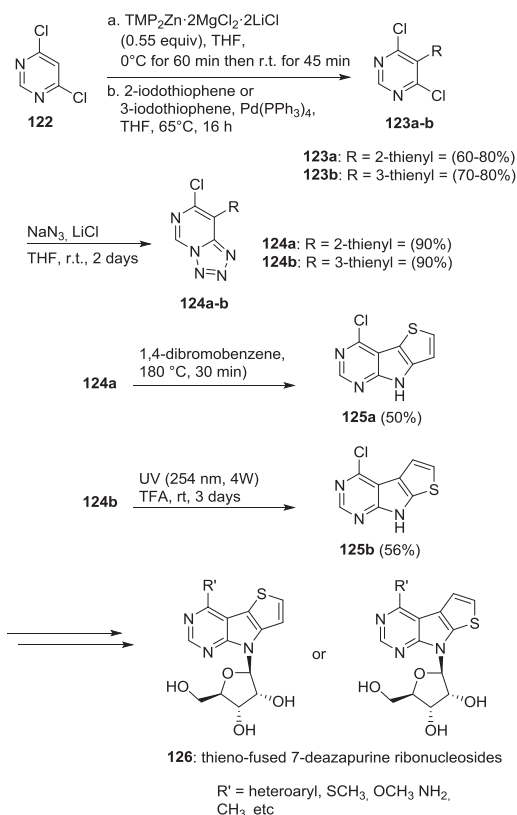
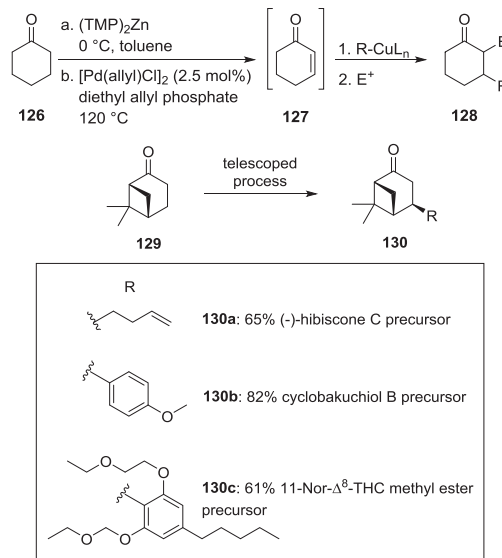
coupling reaction with aryl, phenylvinyl, and phenylalynyl iodonium salts [68]. Here, the TMPZnCl·LiCl base allowed selective metalation of sensitive and electron-deficient (hetero)arenes, from nitrile- and halogen-bearing pyridines to benzoxazoles and caffeine. This was demonstrated by reaction of some substituted-nicotinitriles with iodonium salts, which allowed the synthesis of biologically important precursors, including TRPV1 and AT1 antagonists. Furthermore, a potent inhibitor of the enzyme histone deacetylase (HDAC) [69] was prepared in a few steps by applying this arylation strategy. Deprotonation of 6-bromonicotinonitrile (**117**) with **4** followed by coupling with arylodonium salt **118** afforded arylated product **119** in 57% yield. After a Suzuki cross-coupling with 4-fluorophenylboronic acid, basic hydrolysis of ester **120** and amidation with 1,2-diaminobenzene furnished **121** in 89% yield (Scheme 23).

Nucleosides derived from 7-deazapurines display antineoplastic and antiviral activities. Hoces and co-workers designed a new class of thieno-fused 7-deazapurine ribonucleosides (**126**) with in vitro cytostatic action against a wide spectrum of leukemia and cancer cell lines [70]. Pyrimidine and thiophene rings were linked by initial metalation of 4,6-dichloropyrimidine (**122**) with TMP₂Zn·2MgCl₂·2LiCl under mild conditions, followed by a Negishi cross-coupling reaction with 2- or 3-iodothiophene, which

Scheme 23. Synthesis of HDAC inhibitor **121**.

gave derivatives **123a–b** in 62% and 58% yield, respectively. Nucleophilic substitution with sodium azide gave tetrazolopyrimidines of type **124**, which were further cyclized to thienopyrrolopyrimidines **125a** and **125b** under thermal or photochemical conditions, respectively. Finally, target compounds were obtained after *N*-glycosylation and manipulation of position C-4 of the corresponding purine rings of type **126** (Scheme 24).

α,β -Difunctionalized cyclic ketones are relevant building blocks in organic synthesis, but their preparation often depends on multistep approaches starting from enones [71]. Recently, Newhouse and co-workers developed a versatile one-pot strategy for

Scheme 24. Synthesis of thieno-fused 7-deazapurine ribonucleosides (**125**).Scheme 25. Telescoped process to obtain α,β -difunctionalized ketone **130**.

α,β -difunctionalization of saturated cyclic ketones [72]. In the first step of this process, a zinc enolate was generated by using $\text{Zn}(\text{TMP})_2$, since the lithium bases TMPLi and LiCyan were inefficient. Next, the zinc enolate was oxidized with an allyl-palladium catalyst, to produce an enone intermediate (**127**). This was telescoped with an organocuprate reagent, which allows conjugated addition of a nucleophile at β -position, forming an enolate that could be trapped with an electrophile and giving the α,β -difunctionalized ketones (**128**). This methodology was crucial for the synthesis of some precursors of biologically active compounds such as (-)-hibiscone C (**130a**), cyclobakuchiol B (**130b**), and 11-Nor- Δ^8 -THC methyl ester (**130c**), which was achieved in better yields as compared to previous multistep reports (Scheme 25) [73].

4. Summary and conclusion

Over the last years, the development of new synthetic methods has increased the scope of applications of magnesium- and zinc-TMP amides considerably. These reagents are extremely important in modern synthetic chemistry, as demonstrated through the synthesis of several natural products and pharmacologically relevant targets. $\text{TMPPMgCl} \cdot \text{LiCl}$ is the most frequently used mixed Li/Mg metal base; it can be applied to prepare highly functionalized aromatic and heteroaromatic synthetic intermediates. On the other hand, mixed Li/Mg diamides such as $\text{TMP}_2\text{Mg} \cdot 2\text{LiCl}$ can be employed to metalate of less activated substrates.

Since the first reports on their preparation and reactivity in 2007, $\text{TMP}_2\text{ZnCl} \cdot 2\text{MgCl}_2 \cdot 2\text{LiCl}$ and $\text{TMPZnCl} \cdot \text{LiCl}$ have become an exciting addition to the organic synthesis toolset available to chemists. Aiming to synthesize bioactive targets, synthetic chemists have successfully used these reagents to metalate sensitive heterocyclic substrates.

Alongside reports on the use of LiCl-activated magnesium- and zinc-TMP amides on large-scale building block batch preparations, recent publications have also illustrated the great potential of these reagents in flow chemistry [74]. These new developments will greatly impact the future use of these reagents in organic synthesis.

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